

A background image of an astronaut in a white spacesuit floating in space, with a view of Earth and a nebula. The astronaut's hands are visible, holding a transparent panel.

PROMPTATHON

University of North Carolina – Kenan-Flagler Business School

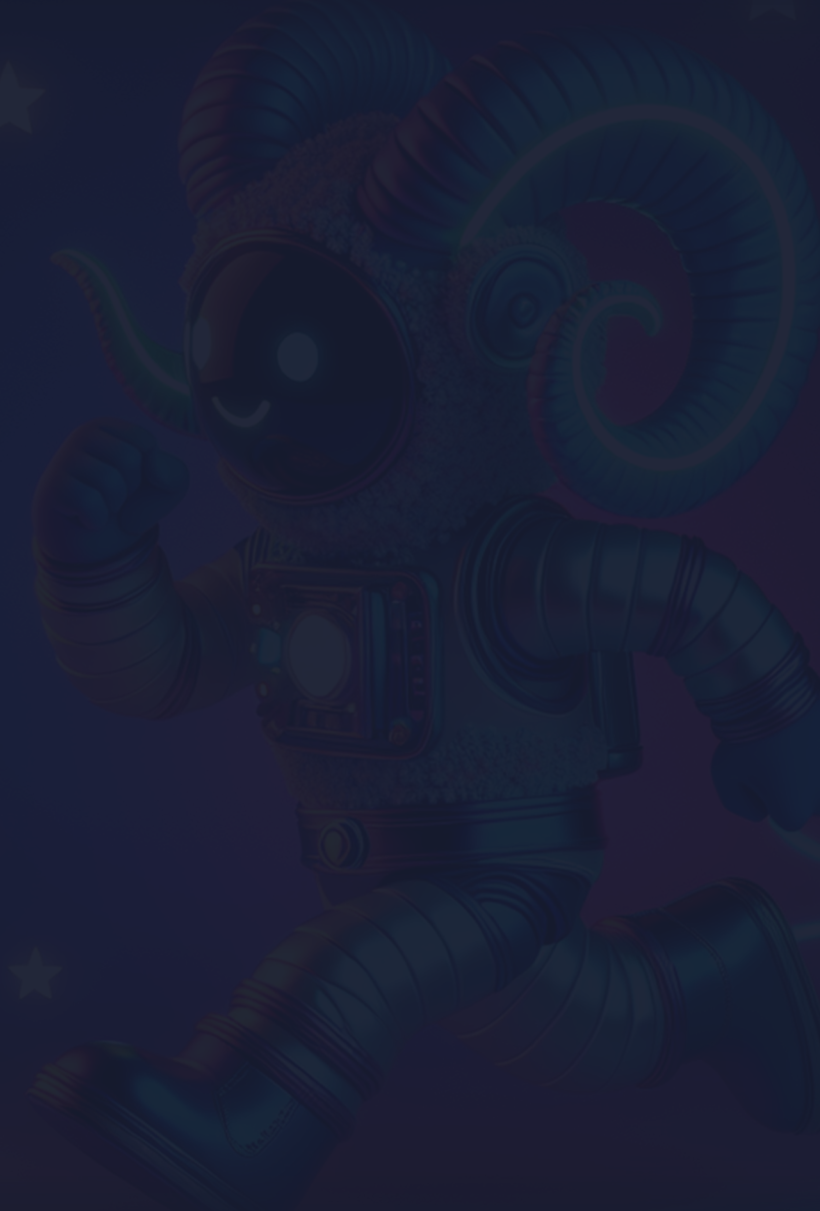
GARRETT BROADY
Senior Technical Trainer

About the Speaker



GARRETT BROADY

- Kinesiology @ University of Maryland (Started w/ CS)
- Trained Elite Athletes for 15 years
- IT > Help Desk > Net Admin for 7 years
- Teaching and Coaching for 4 years > Transition to Technical Training, MCT
- Life-long learner with a passion for making complicated topics easy for all to understand



Session Objectives

01

CHALLENGE EXPLANATION

How do we take a challenge we have and use that to create an Agent that can support or resolve our issue?

02

LEARN ABOUT GEN AI AND AGENTS

What is an Agent? What capabilities exist within nebulaONE®?
How can I build an agent when I'm not an AI expert?

03

BUILD THE AGENT

Take what we have learned, and build out an Agent that aligns with our vision.

Presentation Agenda

01

Understanding GenAI

02

Intro to Agents

03

PACE Framework

04

System Instructions

05

Agent Creation

06

Q&A

So, what are we about to do exactly?

Goal

Research, design, prototype, and pitch a **GenAI Agent that solves a real faculty/staff/student problem, demonstrating clear user value and responsible AI practices.**

Use **BlueChipAI** to ideate, prompt, generate content, and assist in building the Agent and pitch.

Format (3 phases)

- **Phase 1:** Research and planning (30 min)
 - Problem; Persona; Success metric
- **Phase 2:** Design and prototyping (30-60 min)
 - Prompts/workflow; Tools; Build a run-able demo
- **Phase 3:** Iteration and documentation (30 min)
 - Refine; Produce 1-page spec & demo script

What are the reqs?

Rules

- **Teams:** You CAN work in teams (offline – Teams msgs)
- **Data:** Public or synthetic only; no PII or sensitive info
- **Tools:** BlueChipAI for ideation, planning,& prototyping; Personal agent.

Deliverables

- **1-page Agent Spec (PDF/JPG/DOCX)**
 - Agent description, purpose, audience, workflow
- **3-Min Demo**
- **Agent Link** – Share you agent and be sure to click Save after you copy link or else we won't be able to use your agent!

“Judging”

- Problem clarity and evidence
- User value and measurable impact
- Agent design quality and feasibility
- Responsible AI and risk mitigation
- Live demo clarity and reproducibility

Ask BlueChipAI



> What is GenAI?

ELI5

Explain in simple terms

Explain like I'm a College

[Student/Professor/Grad]



What is Gen AI?



GENERATIVE ARTIFICIAL INTELLIGENCE IS A FEW THINGS, REALLY:

Artificial Intelligence is a science that tries to recreate the abilities and capabilities of human intelligence. This deals with decision-making, processing, “thinking” (or reasoning), and executing.



Generative is the process of creating new content based on the knowledge and training of the creator. In this sense, the model is generating content based on its training.



The **large language model** (LLM) that powers genAI can be described as a highly-educated word predictor that has studied billions of conversations and texts to learn how people naturally speak and write.

HOW DOES GEN AI WORK?

- Content
- Machine Learning
- Vectors & Embeddings
- CPUs & GPUs
- NLP & LLMs
- Prompts on Prompts
- Responses on Responses

No, but seriously...

- ✓ LLMs are trained to understand how humans use words and language in different situations and contexts.
- ✓ It takes your "Prompt" and extracts context and semantic meaning
- ✓ Prompt + Context + Semantics + Training = **RESPONSE**

GenAI Formula



PROMPT

The instructions that dictate the style, tone, behavior, and direction the agent should take to generate a response.

CONTEXT

What additional information, knowledge, and reasoning are needed for the model to make its decision and generate content?

SEMANTICS

What is the true meaning behind what is being built by the prompt and context?

TRAINING

The ability to take massive amounts of content and create a usable algorithm to generate a useful response.

Ask BlueChipAI

> Who is the best...?

Context matters!!



What can GenAI do?

Literal Language Processing

- **Structural Understanding:** Sentences, paragraphs, chapters, and complete documents
- **Format Recognition:** Articles, reports, speeches, presentations, and books
- **Technical Content:** Definitions, specifications, documentation, and academic papers
- **Organizational Patterns:** Outlines, lists, tables, and hierarchical information

Conceptual Understanding

- **Abstract Reasoning:** Ideas, theories, hypotheses, and philosophical concepts
- **Emotional Intelligence:** Sentiment analysis, tone detection, and mood interpretation
- **Semantic Relationships:** Context, meaning, implications, and nuanced understanding
- **Literary Devices:** Metaphors, analogies, symbolism, and figurative language
- **Logical Operations:** Inference, deduction, extrapolation, and pattern recognition
- **Cultural Context:** Idioms, references, humor, and social conventions

Visual & Imaginative Language

- **Descriptive Generation:** Vivid descriptions, scene setting, and atmospheric writing
- **Imagery Translation:** Converting visual concepts to text and vice versa
- **Creative Visualization:** Imaginative scenarios, world-building, and conceptual design
- **Sensory Language:** Describing textures, sounds, tastes, and experiences
- **Spatial Understanding:** Directions, layouts, relationships, and dimensional concepts

Transformative Capabilities

- **Style Adaptation:** Adjusting tone, formality, and voice for different audiences
- **Language Translation:** Not just word-for-word, but preserving meaning and cultural context
- **Format Conversion:** Turning prose into bullet points, scripts into narratives, etc.
- **Complexity Adjustment:** Simplifying technical content or elaborating on simple concepts
- **Perspective Shifting:** Rewriting from different viewpoints or for different purposes

Six Pillars of Responsible AI

Fairness

By carefully assessing bias and utilizing diverse data sets, fairness and discrimination mitigation are woven into the foundation of nebulaONE®. nebulaONE furthers this commitment by supporting unbiased model selection and offering transparent configuration options that promote equitable outcomes for all users.

Reliability & Safety

Algorithmic transparency ensures stakeholders understand AI decision-making through clear documentation and explanations, while ethical integrity guides every operation through regular testing and validation to prevent harm and support accurate results.

Privacy & Security

Data is collected ethically, with informed consent and in compliance with regulations, and stored securely using robust encryption and access controls. Data sharing follows strict protocols, ensuring anonymization and customer control within compliant Azure environments.

Inclusiveness

nebulaONE's no-code platform adopts inclusive design principles, empowering users of all backgrounds to build AI agents tailored to diverse needs. To further strengthen inclusiveness, nebulaONE implements user-centric designs and provides clear feedback channels, continually enhancing accessibility for all users.

Transparency

nebulaONE integrates Azure's model interpretability tools, providing stakeholders with comprehensive visualizations and metrics that clarify how AI models reach decisions. Pairing transparent communication with thoughtful design empowers users to understand the underlying processes and outputs of AI systems.

Accountability

Responsibility for each AI system is assigned to designated individuals and teams within nebulaONE. The platform maintains comprehensive audit trails and governance data alongside clear procedures for reporting ethical issues or compliance concerns.

Ask BlueChipAI

> Can you explain Palms please?



Ask BlueChipAI

> Can you explain AI?



Ask BlueChipAI



> First thing that comes to mind when you hear “It’s Raining...”

Ask BlueChipAI



> Tacos!!

Semantic understanding
matters!!

PACE Framework

PROBLEM

What is the problem you're trying to fix –
How would you typically fix this without AI?

ACTION

What's the action or ask that you want the model to complete?

CONTEXT

What additional information is needed for the model to make its decision and complete its output?

EXPECTATIONS

Influence the outcome by presenting examples and expectations for it to match and meet respectively.



PROBLEM

Identify the issue

Example:

I need to improve my time management skills because I am struggling to meet project deadlines.

ACTION

CONTEXT

EXPECTATIONS

PROBLEM



ACTION
Instruct the model

Example:

Generate a list of
effective time
management strategies.

CONTEXT EXPECTATIONS

PROBLEM

ACTION



CONTEXT

Inform the model

Example:

I work remotely and manage multiple projects at once, often collaborating with team members in different time zones.

EXPECTATIONS

PROBLEM

ACTION

CONTEXT



EXPECTATIONS
Influence the outcome

Example:

Provide at least five strategies, with brief explanations for each, and include examples relevant to remote work.

Chain of Thought Prompting



I need to solve a complex problem about [topic]. Let's break this down:

1. First, consider the key factors involved: [list factors].
2. For each factor, analyze how it impacts the situation.
3. Identify potential connections between these factors.
4. Based on this analysis, what conclusions can we draw?
5. Finally, synthesize these insights into a comprehensive solution.

What is it?	Why it works:	When to use it:
This technique involves guiding nebulaONE through step-by-step reasoning processes before arriving at a final response.	• Prompting nebulaONE to generate a chain of thought before answering significantly improves performance on complex reasoning tasks.	• Advanced reasoning tasks. • Accuracy in multistep requests. • When there are multiple ways to complete a task.
	• Forces nebulaONE to reason and connect the dots towards a result	• When process is ultimately important and may cause confusion. • Shortcut to providing complex advanced prompts.

Role-Based Expertise Prompting



I want you to respond as if you are a panel of experts consisting of:

- A leading researcher in [field]
- An experienced practitioner with 20+ years in [industry]
- A critical analyst who identifies potential flaws in business plans
- A creative innovator who thinks outside conventional boundaries

Considering my question about [topic], provide perspectives from each expert, highlighting where they agree and disagree.

What is it?	Why it works:	When to use it:
This technique involves assigning specific expertise roles to the AI to access more specialized knowledge	• This leverages the AI's ability to simulate different cognitive frameworks and provides multiple perspectives on complex issues. "Wisdom of crowds"	• Gaining a fresh perspective • Interdisciplinary projects w/multiple perspectives
	• Different disciplines have unique frameworks, terminology, and approaches to problems. This accesses those frameworks.	• Understanding course material from professor's viewpoint • Interview prep – simulating real life scenarios

Metacognitive Prompting



Please answer this question about [topic]. After providing your answer:

1. Rate your confidence in each part of your response on a scale of 1-10
2. Identify the weakest aspects of your reasoning
3. Suggest what additional information would strengthen your answer
4. Provide alternative perspectives that might challenge your initial response

What is it?	Why it works:	When to use it:
This technique involves asking nebulaONE to reflect on its own reasoning process and evaluate the quality of its response.	• This encourages a self-evaluative process which can potentially identify limitations in its own reasoning.	• Advanced reasoning tasks when solving complex problems. • Evaluating competing theories or interpretations.
	• It provides a way to question assumptions and consider multiple angles.	• Developing research questions and plans. • When seeking to improve and challenge your perspective. • Preparing for discussion-based presentations.

Adversarial Stress Testing



I need information about [topic]. After you provide your initial analysis:

1. Imagine the three strongest counterarguments to your position
2. Respond to each counterargument with evidence
3. Identify any areas where the counterarguments reveal weaknesses in your initial analysis
4. Provide a refined conclusion that incorporates these considerations

What is it?	Why it works:	When to use it:
Challenge the AI's initial response to refine and strengthen it by introducing counterarguments or potential weaknesses.	• Simulating the peer-review process and strengthening through critical evaluation. • Examining how ideas stand up to criticism, you can develop stronger arguments and more nuanced understanding.	• Evaluating a generated idea thoroughly to help identify weaknesses and results. • Preparing arguments and debates. • Strengthening a thesis or stance on a topic.
	• Verifying the process and reasoning methods that nebulaONE takes.	• Testing your understanding of complex course material. • Preparing for presentations and pitches. • Evaluating competing theories and approaches.

Temporal Context Shifting



I want to understand how thinking about [topic] has evolved over time. Please provide:

1. How this would have been understood in [past time period]
2. The current mainstream understanding as of [present year]
3. Emerging perspectives that might become dominant in the next 5-10 years
4. What fundamental assumptions have changed across these time periods

What is it?	Why it works:	When to use it:
Shift the AI's perspective by giving it a vantage point from a specific era or window of time.	• This technique leverages the AI's training on materials from different time periods, providing historical context and highlighting how understanding evolves.	• Studying the evolution of theories, practices, and methods in your field. • Examining how changes might impact your field. • Applying past solutions to current problems.

Persona Framework



For our conversation, I'd like you to adopt the persona of [specific character/expert] with the following characteristics:

- Background: [relevant expertise/experience]
- Communication style: [formal/casual/technical/simplified]
- Perspective: [optimistic/critical/balanced/innovative]
- Special focus on: [particular aspects of topics]

What is it?	Why it works:	When to use it:
Define a specific character with a specific knowledge base, particular expertise, communication style, and perspective.	• This creates consistency in responses and can access specialized knowledge. • Access frameworks that might not be triggered by standard prompting.	• Studying perspectives through the eyes of someone experiencing that situation. • Practice language skills or contextual conversations.
	• Attaches a personality to abstract or unfamiliar concepts, experiences, and perspectives. • Maintains a consistent lens on topics.	• Exploring philosophical, cultural, or ethical positions and perspectives. • Role-playing exercises.

Multi-turn Dialogue Structure



Let's establish a specific conversation structure for exploring [topic]:

1. In the first exchange, you'll help me define the scope and key questions
2. In the second exchange, you'll provide initial analysis and ask clarifying questions
3. In the third exchange, you'll present multiple perspectives or solutions
4. In the fourth exchange, you'll help evaluate options and develop an implementation plan

What is it?	Why it works:	When to use it:
Structured conversation flow with defined phases. The conversation will follow planned stages, where each exchange builds systematically on previous ones.	• Mirrors the structure of effective tutorials and study sessions via scaffolded learning.	• Providing a more thorough analysis where definitive context is needed. • Preparing for exams – Targeted studying. • Developing multi-stage projects and plans.
	• Ensures comprehensive coverage of a topic while maintaining a coherent progression.	• Learning processes that have distinct sequential phases. • Practicing for structured interviews or assessments – Med, CS, "if-then" situations.

Implement Feedback Loops



I want to establish a feedback system for our conversation:

- After each of your responses, I'll rate it on a scale of 1-5.
- I'll provide specific feedback on what was helpful and what could be improved.
- In your subsequent responses, explicitly address how you're incorporating my feedback.
- Periodically summarize the patterns in my feedback to ensure you're adapting correctly
- Present all the feedback in a set of instructions to save and use to prime future conversations.

What is it?	Why it works:	When to use it:
Establish an ongoing refinement process where the AI continuously improves based on your feedback.	• This creates a learning mechanism within the conversation itself, allowing nebulaONE to adapt to your specific needs and preferences.	• Refining written text through multiple drafts. • Working through quiz questions, building a new agent, refining a prompt.
	• Personalized learning and personalized presentation of information.	• Developing or brainstorming projects & ideas that require iteration. • Tailoring audience-specific content • Adjusting explanations to your context.

- An Agent is a genAI tool that can handle a variety of interactions and tasks depending on the sources, capabilities, and instructions that you provide. It's like your personal assistant that can be directed to support you in any way you'd like.
- Agents have evolved from being a general knowledge chatbot to a specific purpose-driven tool that can be as focused or as widely-scoped as you may need.
- Agents get their real power from added components that can be integrated into them - like plugins or modifications, which give an extra boost of knowledge and capabilities to an Agent.

Wait...
what's an —
Agent?

In their simplest form, Agents are a solution to a problem or a tool/skill to enhance the user's experience. What have you known Agents to enhance or resolve?



Job Interview

Agent takes on the persona of an interviewer for a given job role



Quiz Agent

Sends content to a quiz agent that will create a question and respond back to user



Output Agent

Format of the output is changed based on the desired response style, tone, and preference



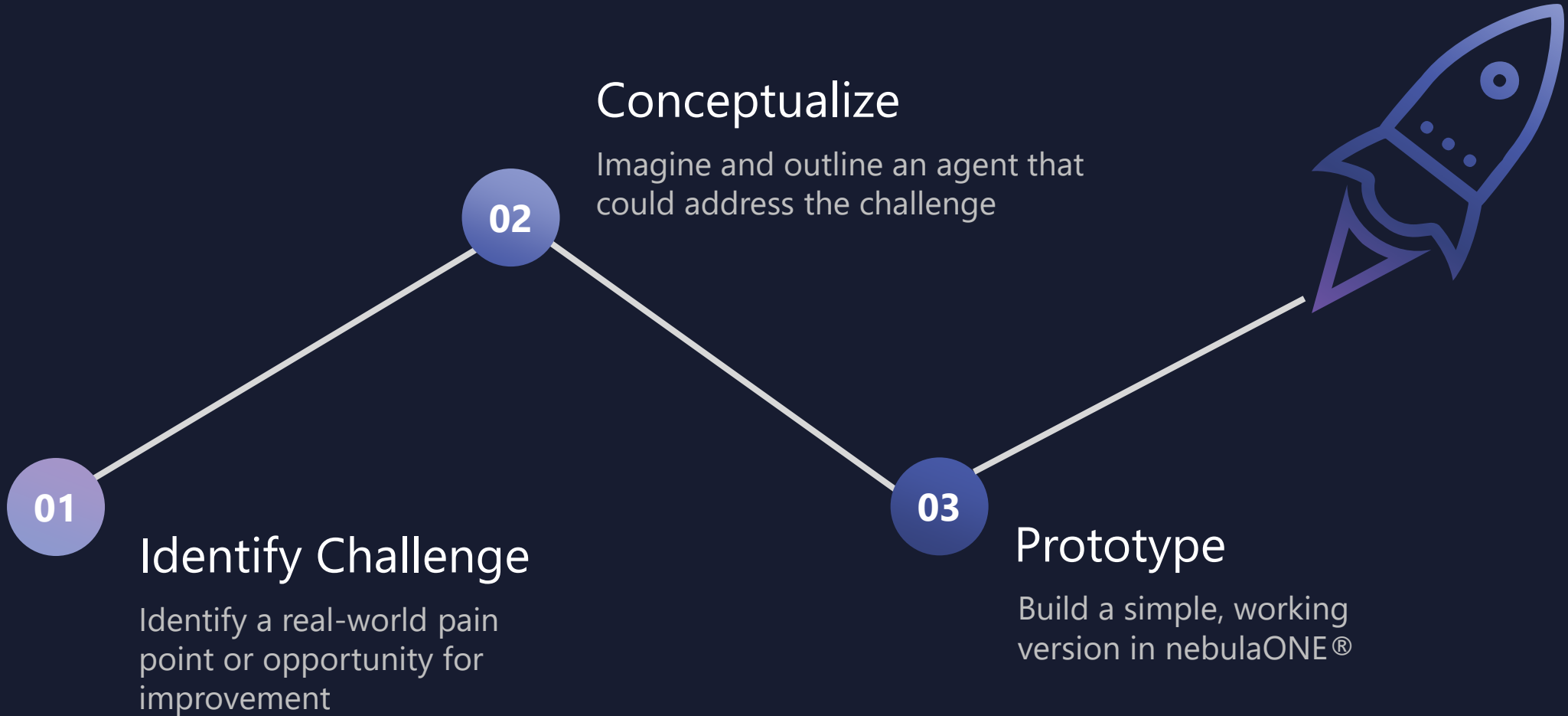
Access to Data

An agent with access and understanding of a dataset

- **Large Language Models:** These are the brains of the operation, commonly referred to as LLMs, SLMs, Fine-tuned models, etc.
- **System Instructions:** Specific guidance on how the Agent should behave, process, and respond in general and in complex situations.
- **Context:** Background information and situational awareness added to the conversation via a user prompt.
- **Knowledge Sources:** Data to help inform the Agent and shape what information it can use when responding to queries.
- **Prompts:** Chat starters to help shape/add specific context for the Agent or a user's input which will directly relate to their desired response/request.

What are common components of Agents?

How do we ideate on a use case?





How do we build out its corresponding Agent?

01

START SIMPLE

- Begin with a straightforward deterministic goal
- Focus on a broad context
- Keep prompts clear and minimal

02

ADD AUDIENCE

- Narrow the scope, context, tone, target
- Add dynamic decision-making and reasoning to further define semantic understanding
- Refine knowledge sources

03

EXTEND CAPABILITIES

- When you have clearly distinct domains or tasks
- Multiple conversation contexts
- Get more advanced with the instructions

Ask BlueChipAI

Let's build an agent



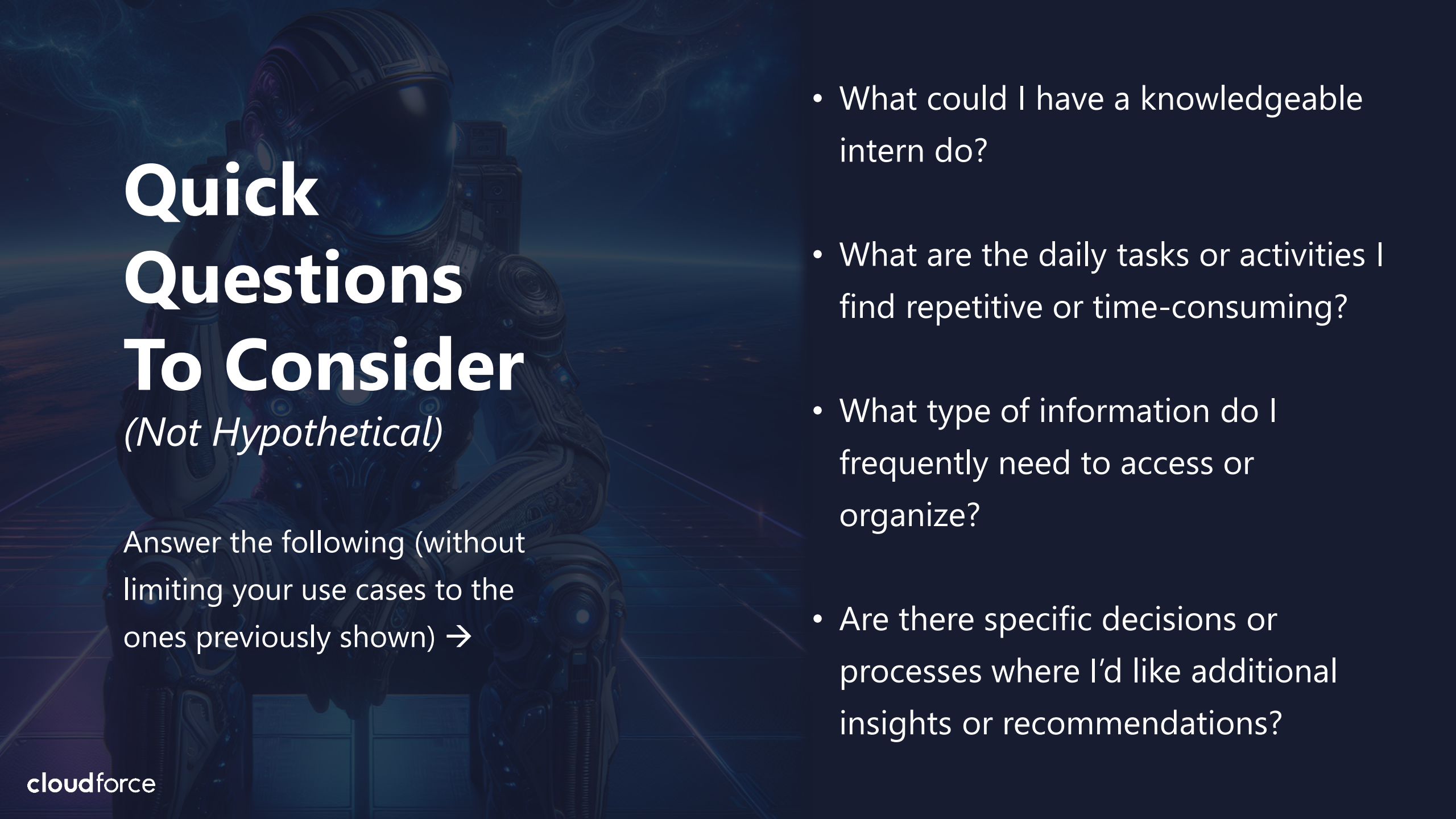
What matters when building an agent?

Knowledge

Audience

Prompts

Access to Data



Quick Questions To Consider

(Not Hypothetical)

Answer the following (without limiting your use cases to the ones previously shown) →

- What could I have a knowledgeable intern do?
- What are the daily tasks or activities I find repetitive or time-consuming?
- What type of information do I frequently need to access or organize?
- Are there specific decisions or processes where I'd like additional insights or recommendations?

The background is a dark blue, stylized illustration. It features two large, symmetrical robotic figures on the left and right sides. These figures have large, curved, horn-like structures on their heads and intricate mechanical details. In the center, between the two large figures, is a smaller, more dynamic figure that appears to be in a combat or jumping pose. The overall aesthetic is futuristic and militaristic.

The BlueChipA Challenge

So, what are we about to do exactly?

Goal

Research, design, prototype, and pitch a **GenAI Agent that solves a real faculty/staff/student problem, demonstrating clear user value and responsible AI practices.**

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“Judging”

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- User value and measurable impact
- Agent design quality and feasibility
- Responsible AI and risk mitigation
- Live demo clarity and reproducibility

Use BlueChipAI

What are —• some examples of agents?

Quick examples to spark ideas

- **Advisor Q&A triage agent**
 - Aim to reduce wait times
- **Syllabus/assignment assistant for students**
 - Goal is to increase comprehension and foundational understanding
- **Policy document agent for staff**
 - Goal is to make it easy for staff to discover and understand policies
- **Faculty research summarizer**
 - Accelerate literature review and find similar research
- **Career services mock interview coach (improve outcomes).**
 - Improve preparation and outcomes based on specific details
- **Case study creation agent**
 - Create mock case studies and AI exercises
- **Course registration optimizer/analyzer**
 - Understand course content and needs (faculty/staff)

TOP ADVANCED PROMPTS TECHNIQUES



MULTI-TURN DIALOGUE STRUCTURE

Setting up a conversation that follows planned stages, where each new exchange builds on what you've already discussed.



TEMPORAL CONTEXT SHIFTING

Changing the time perspective – Like asking “How would someone in 1950 view this?” or “What will this look like in 2040?”



ADVERSARIAL STRESS TESTING

Playing devil's advocate with the AI's ideas to find weak spots and make the final answer stronger.



PERSONA FRAMEWORK

Creating a consistent character for AI to play – Complete with Personality, background, and speaking style throughout the conversation.



ROLE-BASED EXPERTISE

Telling the AI to put on a specific “hat” – Speaking with an expert to frame its knowledge and response style around a specific perspective and language.



FEEDBACK LOOP

Building in chances for you to say, “that’s not quite right” or “exactly what I’m looking for,” so nebulaONE can adjust and improve as you go.



METACOGNITIVE PROMPTING

Asking nebulaONE to double-check its own thinking and consider if there might be better approaches before delivering its final response.

Create Your Personal Agent

1. Finalize your use case.
2. Create your system instructions, using previously shared samples as needed.
3. Upload your knowledge sources, and allow the Agent to suggest possible sources or pinpoint gaps in necessary information.
4. Create starter prompts that align with the anticipated FAQ's of your use case.
5. Test the Agent by analyzing its responses and changing one component at a time until its output is optimal.

FILL IN THE BLANKS

(to begin ideating on the problem you're solving):

"I want to build an Agent that helps **[who]** to **[do what]** by **[using what information]** in a **[what tone]** way, and I'll know it's working when **[success metric]**."

Q & A

**Cloudforce Customer
Success Training
Feedback
Survey**

